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Data Science

Advanced SQL DATE functions.

Table 1 : Employees

--- Q1: Add 6 months to each employee's hire date using
--- DATEADD( )

```
SELECT emp_id,
       name,
       hire_date,
   DATEADD(MONTH, 6, hire_date
FROM Employees;
```

Output table

emp_id	name	hire_date	hire_plus_6_months
1	Alice	2020-01-15	2020-07-15
2	Bob	2021-06-10	2021-12-10
3	Charlie	2023-03-22	2023-09-22

Table 2 : Students

---Q2: Use DATEDIFF() to find age in days from dob to
---today

```
SELECT student_id,
       name,
   DATEDIFF(DAY, dob, CURRENT_DATE) AS age_in_days
FROM Students;
```

Table 3: Events Output table

student_id	name	age_in_days
101	Maya	7 555
102	Ethan	7 807
103	Sienna	7 338

Table 3: Events

--- Q3 : Find how many days are left until each event u

--- DATEDIFF()

```
SELECT event_id
       event_name
   DATEDIFF(DAY, CURRENT_DATE(), event_date)
   AS days_remaining
FROM Events;
```

Output table

event_id	event_name	days_remaining
1	Seminar	-671
2	Workshop	-228
3	Hackathon	-452

Table 4 : Invoices

--- Q4 : Calculate the number of days between issue_date and due_date

```
SELECT invoice_id,
       issue_date,
       due_date,
   DATEDIFF(DAY, issue_date, due_date) AS days_between

FROM Invoices;
```

Output table

invoice_id	issue_date	due_date	days_between
501	2025-03-10	2025-03-25	15
502	2025-04-01	2025-04-15	14
503	2025-04-10	2025-04-20	10

Table 5 : Courses

Q5: Format start_date as 'Month YYYY' using TO_char()

```
SELECT course_id,
       name,
       TO_CHAR(start_date, 'Month YYYY') AS
       formatted_date
FROM Courses
```

OR

```
SELECT course_id,
       name,
   DATE_FORMAT(start_date, 'MMMM.yyyy') AS formatted_date
FROM Courses;
```

Output table

course-id	name	formatted date
201	SQL Basics	May 2025
202	Python	June 2025

TAble 6: Memberships

--- Q6: Create full date from parts using

--- DATE-FROM-PARTS()

```
SELECT member-id,
   DATE_FROM_PARTS(start_year, start_month, start_day)
   AS full_start_date

FROM Memberships;
```

member-id	full-start-date
1	2023-05-10
2	2022-11-25

Table 7: Subscriptions

--- Q7: Extend each renewal-date by 1 year

--- using DATEADD()

```
SELECT sub-id,
       plan,
   DATEADD(YEAR, 1, renewal-date) AS extended_
   renewal_date
FROM Subscriptions;
```

Output table

Sub-id	plan	extended_renewal_date
11	Basic	2026 - 01 - 01
12	Premium	2026 - 03 - 15

Table 8 : Orders

--- Q8 Show current date and difference from order.date
Use CURRENT_DATE() and DATEDIFF()

```
SELECT order_id,
       order_date,
       CURRENT_DATE() AS today_date,
       DATEDIFF(DAY, order_date, CURRENT_DATE()) AS
       days_since_order
FROM Orders;
```

Output table

order_id	order_date	today_date	days_since_order
1001	2025-04-15	2026-04-17	367
1002	2025-04-10	2026-04-17	372

Table 9 : Trainings

--- Q9 : Extract the year from training_date using
--- DATE_PART() or EXTRACT()

```
SELECT training_id
       topic,
  EXTRACT(YEAR FROM training_date) AS
  training_year

FROM Trainings;
```

Output table

training_id	topic	training_year
1	Safety	2025
2	Compliance	2025

Table 10 : Blog_Posts

--- Q10 : Extract hour and minute from published_on.

```
SELECT post_id,
       title,
  EXTRACT(HOUR FROM published_on) AS hour_published,
  EXTRACT(MINUTE FROM published_on) AS minute_published

FROM Blog_Posts
```

Output table

post_id	title	hour_published	minute_published
1	SQL Tips	10	15
2	Data Cleaning	16	45

Table 11: Drivers

--- Q11: Calc days left until license expiry using
--- DATEDIFF() and today's date

```
SELECT driver-id,
       license-expiry,
  DATEDIFF (DAY, CURRENT_DATEC), license-expiry) AS
       days-left
FROM Drivers
```

Output table

driver-id	license-expiry	days-left
301	2025-08-10	-250
302	2023-12-31	-838

Table 12: Messages

--- Display current timestamp and calculate seconds since
--- the message was sent.

```
SELECT message-id,
       sent-timestamp,
  CURRENT_TIMESTAMP AS current_timestamp,
  DATEDIFF (SECOND, sent-timestamp, CURRENT_TIMESTAMP)
  AS seconds-since-sent
```

Output table

message-id	sent-timestamp	Current-timestamp
1	2025-04-19 09-32-45	2026-04-17 06:37:40
2	2025-04-18 23-59-59	2026-04-17 06:37:40

seconds-since-sent
31240295
31276661

Table 13 : Returns

--- Q13 : Add 15 days to return-date using DATEADD()
to show restock-date

```
SELECT return-id,
       return-date,
  DATEADD(DAY, 15, return-date) AS restock-date

FROM Returns;
```

Output table

return-id	return-date	restock-date
901	2025-04-05	2025-04-20
902	2025-04-01	2025-04-16

Table 14 : Assignments

--- Q14 : Convert assigned-on to date using to-date()
(if it is stored as string).

```
SELECT assign-id,
    TO_DATE (assigned_on , 'YYYY-MM-DD' AS
    assigned-on-date

FROM Assignments;
```

Output table

assign-id	assigned-on-date
1	2025-03-01
2	2025-03-05

Table 15 : Meetings

```
--- Q15: Convert scheduled-time to formatted string like '
--- April 19, 2025 at 2:00 pM' using to_char()

SELECT meeting-id,
    TO_CHAR (Scheduled-time, 'Month DD, YYYY "at"
        HH:MI AM') AS formatted_meeting-time

FROM Meetings
```

Output table

meeting_id	formated-meeting-time
1	April 19, 2025 at 2:00 PM
2	April 19, 2025 at 09:30 AM